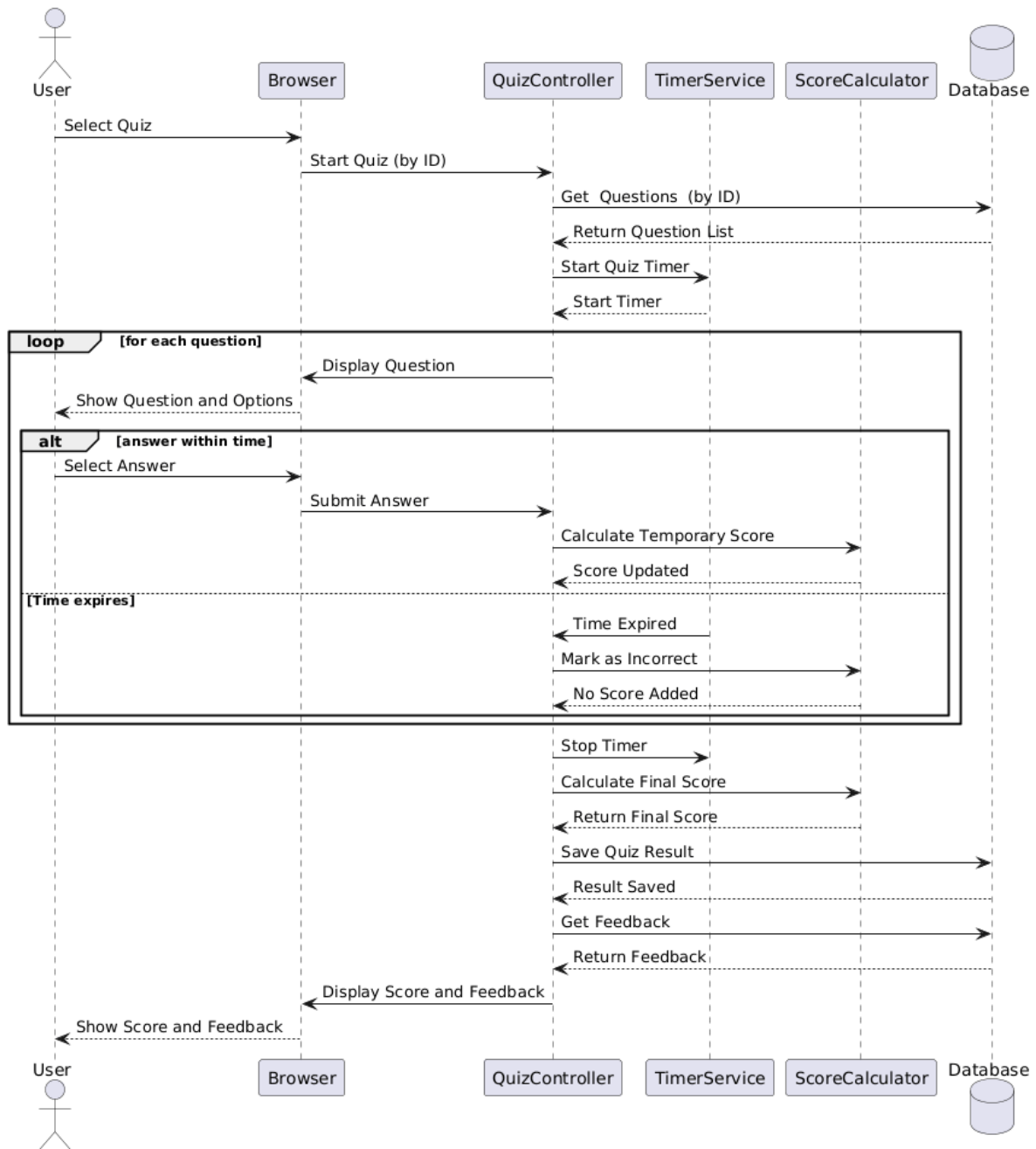




Sequence Diagram Description

Normal Path

- User selects an available quiz in the Browser
- Quiz Controller starts the quiz and loads the questions
- Timer Service starts counting the total quiz time

For each question:

- The browser displays the question and options to the user;
- User selects an answer within the time limit;

- Score Calculator records the partial score;

After all questions:

- Quiz Controller stops the timer and requests the calculation of the final score;
- The result is saved in the Database;
- Quiz Controller requests detailed feedback from the database;
- The database returns the feedback to the Quiz Controller;
- Quiz Controller sends the score and feedback to the Browser;
- The browser displays the final score and detailed feedback to the user.

Alternative Path (Time Expires)

- The timer service detects that the time limit has been reached without the user responding;
- The Timer Service sends a time-expired notification to the Quiz Controller;
- Quiz Controller instructs the Score Calculator to mark the question as incorrect (without adding any score);
- The system automatically advances to the next question;
- The flow continues until the end of the quiz;
- At the end, the User views the results, including unanswered questions marked as incorrect.